Multistability of Elasto-Inertial Two-Dimensional Channel Flow

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Motivation

Polymers in flow: Can reduce or increase drag, cause chaos even without inertia (Elastic Turbulence).

Elasto-Inertial Turbulence (EIT): Puzzling chaotic state where both elasticity and inertia matter.

The Big Question: What triggers and sustains EIT? Two suspects:

1. Viscoelastic Tollmien-Schlichting (TS) waves;
2. A "centre-mode" instability, leading to "arrowhead" flow structures.

Goal: To investigate the dynamical connection between the "arrowhead" structure and EIT.

Miguel Beneitez, et al. Multistability of elasto-inertial two-dimensional channel flow // J. Fluid Mech, 2024.

FENE-P Model

Geometry: 2D channel, periodic in x, walls at $y = \pm 1$.

Incompressibility: $\nabla \cdot \mathbf{u} = 0$;

Momentum: $\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \frac{\beta}{Re}\Delta \mathbf{u} + \frac{1-\beta}{Re}\nabla \cdot \mathbf{T}$;

Polymer Conformation:

$$\partial_t \mathbf{C} + (\mathbf{u} \cdot \nabla)\mathbf{C} + \mathbf{T}(\mathbf{C}) = \mathbf{C} \cdot \nabla \mathbf{u} + (\nabla \mathbf{u})^T \cdot \mathbf{C} + \frac{1}{Re\, Sc}\Delta \mathbf{C};$$

Polymer Stress:

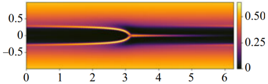
$$\mathbf{T}(\mathbf{C}) = \frac{1}{Wi}\left(f(\mathrm{tr}\,\mathbf{C})\,\mathbf{C} - \mathbf{I}\right), \quad f(x) = \left(1 - \frac{x-3}{L_{max}^2}\right)^{-1}.$$

Key Dimensionless Parameters:
Re — Reynolds number (inertia/viscosity); Wi — Weissenberg number (elasticity/viscosity); β — Solvent viscosity ratio; L_{max} — Maximum polymer extensibility; Sc — Schmidt number (diffusivity).

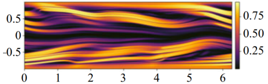
Four Coexisting Attractors

- **Laminar (L):** Steady, smooth flow;
- **Steady Arrowhead Regime (SAR):** Stable, non-chaotic travelling wave with a clear "arrowhead" in polymer stretch;
- **Elasto-Inertial Turbulence (EIT):** Chaotic state. Dominated by active, wavy polymer sheets near the walls;
- **Chaotic Arrowhead Regime (CAR):** Looks like EIT, but with a weak, distorted arrowhead at the center.

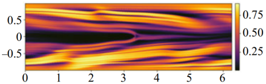
Key Finding: Up to four coexisting attractors (L, SAR, EIT, CAR) for the same parameters! They do NOT transform into one another.



SAR



EIT



CAR

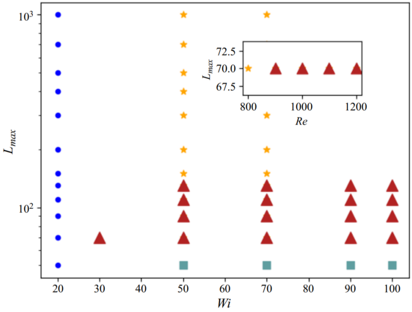
Distinguishing EIT from CAR

Quantitative distinction:

- Gradient at centerline: $C_{grad} = \frac{1}{L_x} \int |\partial_x C'_{kk}(x, y = 0)|\, dx$;
- Projection onto eigenmodes (centre-mode vs. TS-mode).

Phase Diagram (Wi vs. L_{max}):

- Shows regions where different attractors are found;
- Multistability region (red triangles): L, SAR, EIT, CAR all coexist.



Probing State Space with Edge States

Edge State: A saddle state on the boundary between basins of attraction of two attractors.

Method: "Bisection algorithm" to track the boundary (e.g., between EIT and Laminar).

Found Edge States:

1. Between EIT and Laminar: Weakly chaotic state with polymer layers near the wall ($y \approx \pm 0.8$);
2. Between EIT and SAR: Similar weakly chaotic state, but with an arrowhead present.

Insight: These edge states reveal the near-wall polymer layers as the common feature crucial for sustaining chaos.

Energy Transfer: The Engine of Chaos

Key Process: Energy transfer from fluid motion (kinetic energy) to polymer stretch (elastic energy).

Energy Flux:

$$\Pi'_e = \frac{1-\beta}{Re} T'_{ij} S'_{ij},$$

where T'_{ij} and S'_{ij} are perturbative polymer stress and strain rate.

Spatial Location: Analysis of Π'_e shows the maximal energy transfer occurs at $y \approx \pm[0.75, 0.85]$ — exactly where the polymer layers are in the edge states!

Conclusion: The self-sustaining mechanism of EIT (and CAR) is driven by near-wall activity, not by the central arrowhead.

Conclusions

1. **Multistability:** Discovered up to four coexisting attractors (L, SAR, EIT, CAR) in 2D viscoelastic channel flow;
2. **Arrowhead is "Benign":** The "centre-mode" arrowhead structure (SAR) is stable and not dynamically connected to the onset of EIT in 2D. It can sit on top of chaos (CAR) but doesn't drive it;
3. **Engine of Chaos is Near-Wall:** The self-sustaining mechanism for EIT/CAR is an identical near-wall process, involving energy transfer at $y \approx \pm 0.8$.


THANKS
FOR YOUR
ATTENTION